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N + - Network Fundamentals & Building Networks.

Configure IPv6 Addressing.

TASK 1.

- ❖ Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

➤ Assigning a Static IPv6 Address:

System Used: Windows

⌘ Assign a Static IPv6 Address

Open **Command Prompt as Administrator** and run:

```
netsh interface ipv6 set address name="Ethernet" address=2001:db8:1::10
```

⌘ 2. Verify the Configuration

Run:

```
Ipconfig
```

⌘ The output should show:

```
IPv6 Address: 2001:db8:1::10
```

Result: The static IPv6 address **2001:db8:1::10** has been successfully assigned and verified.

TASK 2.

- ❖ **Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address, Use the ping6 or ping -6 command for IPv6 testing.**

➤ **Unique Local IPv6 Configuration:**

System Used: Windows

⌘ **Configure a Unique Local IPv6 Address**

Open **Command Prompt as Administrator** and run:

```
netsh interface ipv6 set address name="Ethernet" address=fd00:1::10
```

Here, **fd00:1::10** is a Unique Local IPv6 address because it starts with **fd**.

⌘ **Test Connectivity**

Assume the other LAN device has the address **fd00:1::20**.

Run:

```
ping -6 fd00:1::20
```

Result: Successful replies confirm that both devices can communicate using their Unique Local IPv6 addresses.

TASK 3.

- ❖ **Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect, Document the steps and the observed address behavior in a text file.**

➤ Temporary IPv6 Address:

System Used: Windows

⌘ **Check the Current IPv6 Address**

Open **Command Prompt** and run:

Ipconfig

Note down the current IPv6 address of the network adapter.

⌘ **Disconnect and Reconnect the Network**

Disconnect the Wi-Fi/Ethernet connection and reconnect it. Then run:

Ipconfig

again and note the IPv6 address.

⌘ **Restart the System**

Restart the computer and run:

Ipconfig

again to check the IPv6 address.

Observation:

After reconnecting the network or restarting the system, the temporary IPv6

address may change. This shows that a temporary IPv6 address is not permanently assigned to the device.

Result:

The IPv6 address was checked before and after reconnection/reboot, and its changing behavior was documented.

TASK 4.

❖ **Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.**

- **Zomato-Style Restaurant Search:**
Python Script:

```
restaurants = {  
    "Dominos Pizza": "2001:db8:1::10",  
    "Burger King": "2001:db8:1::20",  
    "Subway": "2001:db8:1::30",  
    "Biryani Blues": "2001:db8:1::40"  
}  
  
search = input("Enter restaurant name: ").lower()  
  
for name, ipv6 in restaurants.items():  
    if search in name.lower():  
        print("Restaurant:", name)  
        print("IPv6 Address:", ipv6)
```

Example Output:

Enter restaurant name: burger
Restaurant: Burger King
IPv6 Address: 2001:db8:1::20

TASK 5.

❖ Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

➤ IPv6 Subnetting and Manual Configuration: AI-Generated IPv6 Subnetting Scenario

A small food delivery startup uses the IPv6 network **2001:db8:100::/48**. It is divided into three subnets for different departments:

<u>Department</u>	<u>Subnet</u>
Delivery Partners	2001:db8:100:1::/64
Restaurants	2001:db8:100:2::/64
Admins	2001:db8:100:3::/64

Configuration Steps (Windows)

To manually assign an address from the **Delivery Partners** subnet, open Command Prompt as Administrator and run:

```
netsh interface ipv6 set address name="Ethernet"  
address=2001:db8:100:1::10
```

Verification:

```
Ipconfig
```

The IPv6 address **2001:db8:100:1::10** should appear under the Ethernet adapter.

Result: A valid IPv6 address from the selected subnet was manually assigned and verified.